Rotation-Invariant Knowledge Distillation for Remote Sensing Object Detection

Feiyi Li, Xiao Zhang, Wenda Zhao, *Member, IEEE*, Haipeng Wang, and You He

***Abstract*—Detecting small-rotated objects in remote sensing remains a challenging task due to feature dilution and insufficient rotation invariance. Feature dilution arises when small object features are overwhelmed by background noise and progressively lost as network depth increases. Meanwhile, the lack of rotation invariance stems from the fixed nature of convolution, which struggles to handle arbitrary orientations. To address these challenges, we propose a rotation-invariant knowledge distillation (KD), a visual-language models (VLMs) driven KD framework tailored for optimizing small-rotated object detection in remote sensing. Our method introduces two novel components: enhanced-consistency feature distillation (ECFD) and rotation-invariant feature distillation (RIFD). ECFD mitigates feature dilution by aligning consistent language representations from VLMs with cross-depth features, ensuring consistent small-rotated object representation across different depths. RIFD enhances rotation invariance by leveraging VLMs to distill robust rotational knowledge into detectors, aligning positive and negative language features with detector features to reduce sensitivity to orientation changes and mitigate class confusion. Without introducing additional computational overhead during inference, our method significantly improves the performance of remote sensing object detectors. Extensive experiments on public remote sensing datasets with complex scenes demonstrate the state-of-the-art results. The code is available at https://github.com/Shower-Lee9527/CRKD**

***Index Terms*—Knowledge distillation (KD), remote sensing object detection (RSOD), small-rotated objects, visual-language models (VLMs).**

I. Introduction

REMOTE sensing object detection (RSOD) [1] is a fundamental component of visual perception in complex remote sensing scenarios. Recent advances in deep learning have significantly boosted research in this field, driving growing scholarly interest. Some studies in RSOD focus on addressing the challenges of detecting small objects under extremely low pixels [2], [3], while others explore methods for handling rotated objects with arbitrary rotation angles [4], [5]. However, in real-world scenarios, numerous objects are characterized by the dual challenges of being both small in scale and arbitrary in orientation, rather than exhibiting just one of the above characteristics [6], [7]. Optimizing small-rotated object detection across various detectors remains an unexplored challenge for real-world applications.

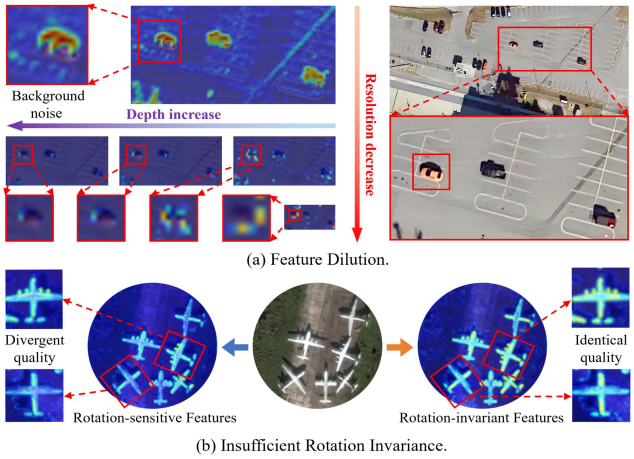


Fig. 1. Primary causes of false positives and false negatives. (a) Background noise introduces redundant activation features on the edges of SVs. The redundant features lack object-specific information and inevitably interfere with semantic representations. As depth increases and resolution decreases, the semantic information of SVs progressively degrades. (b) (Left) Features extracted by the original CNN. (Right) Features from distilled CNN. In contrast to the identical quality of the features on the right, the features extracted by the original CNN exhibit significant variation for airplanes at different angles. This misalignment in feature quality within the same category significantly hampers detection performance.

Detecting small-rotated objects poses significant challenges, especially in reducing false positives and false negatives. These issues are predominantly attributed to two factors listed as follows.

1) Feature Dilution: Small-rotated object features are susceptible to degradation or loss in deep networks due to interference from background noise [1], [8]. As the depth of the network increases, the features progressively shift from focusing on texture details to emphasizing higher level semantic information. Concurrently, the resolution of these features decreases, inevitably replacing detailed local patterns with broader contextual representations. Small-rotated objects occupy only a few pixels, which limits their contribution to deep feature representations dominated by global context. This

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leads to the dilution of discriminative features for small-rotated objects, as they are progressively overshadowed by prominent noise from the background [see Fig. 1(a)]. Consequently, feature dilution hampers the network's ability to localize and classify small objects, diminishing detection accuracy in scenarios with significant scale variation.

2) Insufficient Rotation Invariance: Convolutional neural networks (CNNs) exhibit poor rotation invariance [9] due to convolutional filters not designed for capturing rotational features. This angle dependence in feature extraction causes similar objects to yield distinct representations under rotation variations [see Fig. 1(b)]. Consequently, insufficient rotation invariance produces low-quality features that severely degrade category classification and bounding box regression performance for small-rotated objects under arbitrary orientations. While existing methods [9], [10] attempt to enhance rotation invariance through hand-crafted rotation-equivariant components, their reliance on predefined rotation priors severely limits generalization to arbitrary orientations in remote sensing scenarios. To overcome these limitations, we turn to visual-language models (VLMs), which offer a fundamentally different approach to achieving rotation invariance through language-guided representations.

VLMs [11], [12], benefiting from training on large-scale, loosely aligned image–text pairs, have demonstrated exceptional capabilities in multimodal representation learning. While a small-rotated object may exhibit arbitrary orientations and occupy limited pixels, a language representation can capture the core content of the category in a highly concise and complementary manner. Such language representations are naturally consistent and rotation-invariant, remaining unaffected by changes in orientation or network depth. This motivates recent progress in language-guided object detection, such as PromptDet [13] and VLDet [14], which demonstrate the benefits of integrating VLMs into detection pipelines by aligning visual features with textual prompts. Inspired by these advances, we propose a VLMs-driven knowledge distillation (KD) framework, termed **CRKD**, which transfers the consistent and rotation-invariant language semantics of VLMs to small-rotated remote sensing object detectors.

Building on the previous analysis, feature dilution of small objects arises from background noise and network depth. This limitation renders effective distillation within a single depth particularly challenging. In this regard, we propose enhanced-consistency feature distillation (ECFD) to enhance small object features across multiple depths, mitigating feature dilution. ECFD aligns the consistent language features generated by the VLM using prompt templates with the cross-depth features of the detector. The distillation is applied across features of the same object at different depths, ensuring that small object features do not degrade across depths. Moreover, the lack of rotation invariance in features arises from the sensitivity of convolutional layers to rotation angles. This insufficiency causes detector features of similar classes to depend more on rotation angles than on intrinsic class differences. We introduce rotation-invariant feature distillation (RIFD) to distill rotation invariance from VLMs to detectors. RIFD strategically aligns linguistic class representations with visual instance features at a consistent depth, enhancing the rotation invariance via multimodal projection and distillation. Such distillation enhances the model's ability to handle rotational variations, thereby improving resistance against angular transformations.

Our approach offers a practical solution for RSOD applications and introduces a novel, consistent, and rotation-invariant KD mechanism. We evaluate our method on two challenging benchmarks, showing its superior performance in detecting small-rotated objects. In addition, experiments on both single-stage and two-stage detectors demonstrate the generalizability of our method.

In summary, our contributions are as follows.

1) We propose CRKD, the first KD framework specifically designed to optimize small-rotated objects in RSOD. Without introducing additional modules during inference, CRKD significantly improves the performance of RSOD models for these challenging objects.
2) We introduce ECFD, a module that mitigates small object feature dilution due to background noise and network depth. By aligning consistent language prior with cross-depth detector features, ECFD ensures consistent feature representation across depths.
3) We introduce RIFD, a module that enhances rotation invariance by distilling knowledge from VLMs. It aligns positive and negative language features with detector features, improving resilience to rotational variations and reducing class confusion.

II. Related Work

A. Oriented Object Detection

A series of oriented object detectors have been developed to handle dense layouts and arbitrary orientations in remote sensing imagery [4], [5], [15], [16], [17]. RoI Transformer [15] converts horizontal RoIs into rotated RoIs for spatially invariant feature extraction, while Oriented R-CNN [16] introduces rotation-parameterized proposals and rotated RoI alignment for efficient detection. R^3Det and S^2ANet [4], [5] refine multiscale rotated anchors and adaptively adjust angles and shapes. SASM [17] proposes dynamic sampling and evaluation strategies applicable to diverse frameworks. Despite these advances, most approaches still fail to address the intrinsic rotation sensitivity of CNNs. Representative methods such as ReDet [9], rotation-invariant convolutions based on cyclic polar coordinates [18], coordinate-based designs [19], and RIFormer [20] essentially rely on hand-crafted mathematical transformations (e.g., Fourier transforms and rotation-specific filters). These designs depend on explicit priors about rotation properties, including fixed-angle partitions or periodic constraints, which restrict their generalization to arbitrary orientations in complex remote sensing scenes. On the other hand, methods like adaptive rotated convolution [21] and UAV rotation generalization [22] focus on preserving rotation invariance during the RoI feature extraction process. However, since the underlying features are already degraded by background noise and rotation sensitivity, improving only the RoI transformation step cannot fundamentally resolve the problem.

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In contrast, our method bypasses hand-crafted priors and RoI-specific adjustments by distilling consistent, rotation-invariant semantics from VLMs into detectors, thereby strengthening feature representations across the entire network and enabling more accurate detection of small-rotated objects.

B. Small Object Detection

Small objects [23] are often overwhelmed by background clutter, making accurate recognition and localization highly challenging. Early efforts enhance resolution directly, with Deng et al. [24] employing deconvolution and subpixel convolution to recover high-resolution features, and Pan et al. [25] introducing guided upsampling to generate finer representations. GAN-based methods [26], [27] attempt to hallucinate detailed structures, but often introduce artifacts and increase inference cost [28]. To avoid such drawbacks, alternative designs emphasize efficient strategies such as data augmentation [29], scale-aware training [30], [31], or contextual modeling [32], [33]. In remote sensing, SCRDet [6] and SCRDet++ [7] explicitly enhance small-rotated object detection by combining attention mechanisms and denoising strategies. Our work follows a different route by exploiting VLM-based KD, which strengthens small object features while keeping inference efficient.

C. KD for Object Detection

KD [34] transfers supervision from a teacher model to a student, with early works focusing on logits [35] or bounding boxes [36]. Feature-level KD has also gained traction: Zhu et al. [37] introduce ScaleKD to refine multiscale feature alignment, while Guo et al. [38] and Li et al. [39] highlight the importance of foreground separation in distillation. In remote sensing, Qi et al. [40] design scale-specific loss functions for multiscale feature transfer, and Yang et al. [41] use auxiliary losses to enforce multiorientation feature learning. However, these approaches are still hindered by background noise and rotation sensitivity, limiting their effectiveness for small-rotated objects. Inspired by ScaleKD [37], we propose cross-depth foreground/background augmentation and cross-instance decoupling with unified VLM features to robustly address small-object challenges in remote sensing.

D. VLM-Based Object Detection

Recent advances in VLMs have spurred a new wave of multimodal detection frameworks. GLIP [42] aligns detector features with text embeddings for open-vocabulary recognition, while DetCLIP [43] and FIBER [44] extend this idea to region-level alignment. Grounding DINO [45] further demonstrates the benefit of grounding VLM knowledge into large-scale detectors. Co-DETR [46] introduces a collaborative hybrid assignment scheme that strengthens feature learning in DETR by leveraging multiple label assignment strategies without adding inference cost. VLDet [14] formulates object-language alignment as a set matching problem, directly training open-vocabulary detectors from image–text pairs. While these methods demonstrate the effectiveness of VLM guidance in bridging vision and language, they primarily aim to improve open-vocabulary generalization and class transferability. In contrast, our work exploits VLMs in a fundamentally different way: instead of expanding the category space, we distill consistent and rotation-invariant semantics from VLMs into detectors.

III. Method

A. Preliminary

In this section, we briefly review the Oriented RPN and Rotated RoIAlign [16]. Building upon these foundational techniques, we introduce scale-aware feature alignment as a preliminary step toward developing our distillation method. Subsequently, we provide a detailed explanation of the proposed distillation approaches in Section III-B.

The Oriented RPN and Rotated RoIAlign are two components proposed in Oriented R-CNN [16] and RoI Transformer [15] for oriented object detection. Specifically, Oriented RPN takes the five-level FPN features $\{P_2, P_3, P_4, P_5, P_6\}$ as input and produces coarse proposals at four scales from P_2 to P_5. Each oriented proposal is represented as a 6-D vector $\rho = (x, y, w, h, \Delta\alpha, \Delta\beta)$, where (x, y) is the center coordinate, w and h represent the width and height of the external horizontal bounding box, and $(\Delta\alpha, \Delta\beta)$ denote the offset distances. By offsetting the vertices from the center of the external bounding box, Oriented RPN accurately predicts the oriented proposals, with the guidance of the offset vector $(\Delta\alpha, \Delta\beta)$. Rotated RoIAlign then converts these proposals from oriented parallelogram boxes to oriented rectangular boxes through a simple diagonal stretching operation. The oriented rectangular proposals obtained at different scales are then projected onto the corresponding FPN feature maps. After partitioning each proposal into $m \times m$ grids, RoIAlign is applied to extract fixed-size RoI features. For a proposal ρ, we compute the rotated RoI feature $F^{\rho}_{P_i}$ at level $P_i \in \{P_2, P_3, P_4, P_5, P_6\}$ and grid index (u, v) as

$$F^{\rho}_{P_i}(u, v) = \frac{1}{n} \sum_{\rho(x,y) \in \text{area}(u,v)} F_{P_i}\Big(R\big(\text{stretch}(\rho)\big)\Big) \tag{1}$$

where F_{P_i} is the feature map of level P_i output by the FPN, n is the number of pixels within area(u, v), stretch$(\cdot)$ denotes the diagonal stretching operation, and $R(\cdot)$ represents the rotation transformation [16].

1) Scale-Aware Feature Alignment: As discussed above, the cross-depth detector features act as a bridge in the distillation process. Effectively extracting and aligning them with unified language features is critical to distillation performance. While cross-depth instance-level detector features can be readily obtained using Oriented RPN and Rotated RoIAlign, aligning these features with unified language features is challenging due to the shape mismatch between the features produced by Rotated RoIAlign and the visual features in the VLM. This requires the addition of adapters at each level, greatly complicating the distillation process. Furthermore, the powerful ability of the visual encoder in the VLM to extract text-aware visual features is entirely overlooked.

To address these challenges, we introduce *scale-aware feature alignment* as a preliminary step in the distillation

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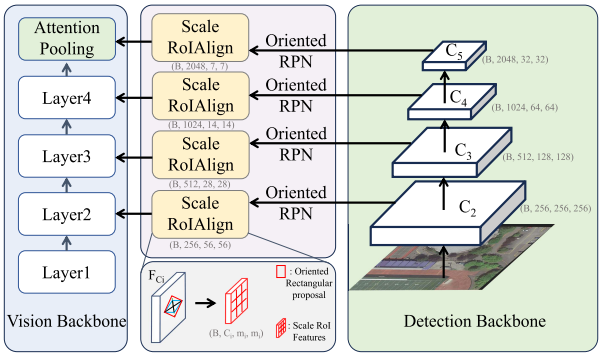


Fig. 2. Pipeline of scale-aware feature alignment for obtaining distillation bridge features that are aligned with unified language features. It comprises two components as follows. 1) Oriented RPN, which generates proposals $P \in \mathbb{R}^{N\times 6}$ from multiscale backbone features. Here, N represents the maximum number of proposals generated per image, and 6 denotes the parameters defining each proposal. 2) Scale RoIAlign, which is used to produce scaled-rotated RoI features $F^{\rho}_{C_i}$.

process, as illustrated in Fig. 2. Specifically, this module comprises the original Oriented RPN and the Scale RoIAlign, an enhanced version of the Rotated RoIAlign. Our motivation is to harness the visual encoder in VLM to extract distillation bridge features that naturally align with unified language features, while maintaining a minimal and efficient structural design. To achieve this, we replace the FPN features with backbone features as input to the module, avoiding the need for additional adapters to align channel dimensions. Moreover, for proposals from different levels, Scale RoIAlign divides them into grids corresponding to the feature map sizes at the corresponding levels in the VLM's vision backbone. RoIAlign is then performed on these grids to extract features, ensuring a structured and scale-consistent representation. For a given proposal ρ, the scaled-rotated RoI feature $F^{\rho}_{C_i}$ is computed in the same manner as $F^{\rho}_{P_i}$. However, unlike $F^{\rho}_{P_i} \in \mathbb{R}^{B\times c\times m\times m}$, the feature output by Scale RoIAlign, $F^{\rho}_{C_i}$, is represented as $\mathbb{R}^{B\times c_i\times m_i\times m_i}$, where c_i and m_i are computed as follows:

$$c_i = c \times 2^{i-2} \tag{2}$$

$$m_i = m \times 2^{5-i} \tag{3}$$

where c denotes the number of channels in the FPN output features (default: 256), m represents the grid size of the output from Rotated RoIAlign (default: 7), and i corresponds to the feature level from which the proposal is generated.

Since $F^{\rho}_{C_i}$ is now structured to match the shape of the output features from the ith layer of the vision backbone in the VLM, it can be seamlessly fed into the subsequent layers of the vision backbone. This enables a progressive transformation, ultimately yielding bridge features F^{ρ}_{V} that are naturally aligned with unified language features, computed as

$$F^{\rho}_{V} = \mathrm{Attn}\left(\mathcal{B}_i\left(F^{\rho}_{C_i}\right)\right) \tag{4}$$

where $\mathcal{B}(\cdot)$ denotes the feature propagation through the subsequent layers of the vision backbone in VLM, starting from the ith layer, and Attn$(\cdot)$ denotes the attention pooling mechanism that enhances the alignment with unified language features.

B. Overview

Fig. 3 presents an overview of CRKD, illustrating its key modules as follows: 1) a *scale-aware feature alignment* module, which generates distillation bridge features aligned with unified language features; 2) an *ECFD* module, which mitigates small object feature dilution by transferring knowledge from unified language features to detector features at multiple depths; and 3) an *RIFD* module, which addresses the adverse effects of varying rotation angles on feature discrimination by aligning cross-instance detector features with unified positive/negative language features. Specifically, the *scale-aware feature alignment* module has been thoroughly discussed in the preliminary. The two KD modules, which form the core of our approach, are detailed in Sections III-C and III-D. Notably, our method distills features solely from the detector's backbone, ensuring plug-and-play compatibility across diverse detector architectures without being affected by the neck or subsequent components. Consequently, these parts are omitted in the following descriptions.

To facilitate the introduction of our method, we represent an oriented object detection dataset as $\mathcal{D} = \{(I_i, \mathcal{O}_i)\}_{i=1}^{|\mathcal{D}|}$, where I_i is the ith image and $\mathcal{O}_i = \{(b_{\mathrm{ij}}, c_{\mathrm{ij}})\}_{j=1}^{|\mathcal{O}_i|}$ is the set of annotations corresponding to the objects. Each annotation consists of an oriented bounding box $b \in \mathbb{R}^5$ and a class label $c \in \mathcal{C}$, where $\mathcal{C}$ denotes the set of all possible object classes. For simplicity, we omit the tokenization step and introduce the CLIP [12] text encoder as $\mathcal{T}$ to extract unified language features. The CLIP visual encoder is denoted as $\mathcal{E}$, which processes the features output by the scale-aware feature alignment module.

C. Enhanced-Consistency Feature Distillation

Based on the previous analysis, the primary cause of missed small objects is feature dilution, which fundamentally arises from persistent background noise interference in deep networks. Preprocessing with upsampling operations or pseudo-feature generation before training often leads to artifacts [28], making it challenging to preserve feature integrity. Moreover, incorporating additional feature fusion modules, anchor sampling strategies, or explicit denoising modules introduces substantial computational overhead and necessitates carefully designed loss functions, which may struggle to generalize to extreme scenarios such as low contrast or densely distributed small objects.

The unified language feature, derived from visual-language space alignment, inherently avoids artifact issues and does not require manually designed loss functions. In addition, its plug-and-play design incurs no extra computational cost. Building on this insight, our proposed ECFD mitigates persistent background noise interference in deep networks by aligning unified language features with cross-depth detector features. Specifically, we first construct a set of prompt templates $\mathcal{P}_{\mathrm{fg}}$ that explicitly describe remote sensing scenes. Each template is combined with category names to generate corresponding textual descriptions. For an instance belonging to category $c \in C$, its textual description is generated through the following process:

$$t_c = p.\mathrm{replace}\left([v_c], c\right), \quad p \sim \mathcal{U}\left(\mathcal{P}_{\mathrm{fg}}\right) \tag{5}$$

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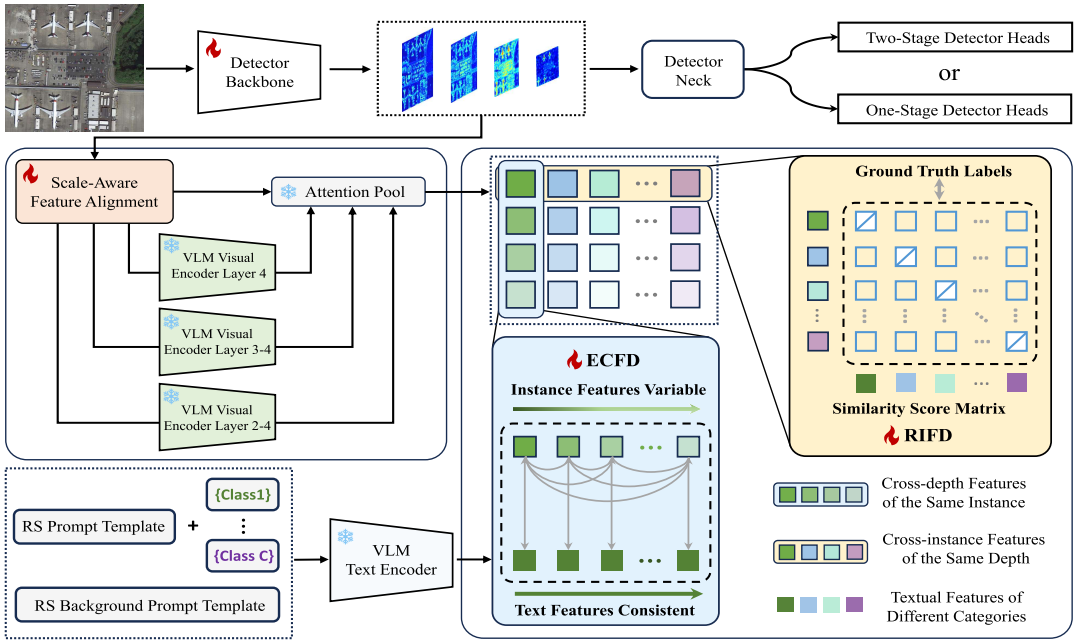


Fig. 3. Pipeline of our proposed framework. Given an input image, the detector first extracts distillation bridge features via the backbone and scale-aware feature alignment. The ECFD module aligns these bridge features with disentangled foreground/background unified language features to suppress background noise-induced feature dilution. The RIFD module formulates a rotation-invariant distillation matrix $\mathcal{M}(\mathbf{v}^d, \mathbf{z})$, enforcing class-specific (angle-agnostic) feature learning through a distillation loss constrained to a single depth layer. Training integrates ECFD and RIFD losses to refine features, ensuring robust performance on small-rotated objects. Notably, these modules possess plug-and-play properties, enabling seamless integration into existing detectors and incurring no additional computational cost during inference.

where p denotes a template uniformly sampled from $\mathcal{P}_{\text{fg}}$, and $[v_c]$ represents the placeholder token for the category name within the template structure. Similarly, we construct a complementary set of background prompt templates $\mathcal{P}_{\text{bg}}$ explicitly designed to characterize background information in remote sensing scenes. Since background templates lack category placeholders, we directly sample $p_{\text{bg}} \sim \mathcal{U}(\mathcal{P}_{\text{bg}})$ to obtain the textual description t_{bg} for background instances. The unified language features are computed as

$$\mathbf{z} = \mathcal{T}\left(\bigoplus_{c\in\mathcal{C}} \tilde{t}_c \oplus \tilde{t}_{\text{bg}}\right) \tag{6}$$

where $\tilde{t}_c$ and $\tilde{t}_{\text{bg}}$ denote tokenized textual representations of foreground and background descriptions, respectively. The operator $\oplus$ represents token concatenation, and $\bigoplus_{c\in\mathcal{C}} \tilde{t}_c$ denotes the concatenation of foreground textual tokens across all categories in $\mathcal{C}$. This formulation explicitly disentangles foreground semantics from background context, enabling robust background noise suppression in deep networks.

Through scale-aware feature alignment, we transform cross-depth detector features into distillation bridge features that are structurally aligned with unified language features. For a given proposal ρ of category c, the corresponding distillation bridge features $\mathbf{v}^\rho$ are defined as

$$\mathbf{v}^\rho = \left\{\mathbf{v}_2^\rho, \mathbf{v}_3^\rho, \mathbf{v}_4^\rho, \mathbf{v}_5^\rho\right\} \tag{7}$$

where each $\mathbf{v}_i^\rho$ is computed following (4), with the subscript i indicating that the feature is extracted from the C_i layer of the detector. Then, we define the optimization objective of ECFD for a given proposal ρ as follows:

$$l(\mathbf{v}^\rho, \mathbf{z}) = \begin{cases} \displaystyle\sum_{l=2}^{L} \frac{\mathbf{v}_l^\rho \cdot \mathbf{z}_c^T}{\|\mathbf{v}_l^\rho\| \cdot \|\mathbf{z}_c\|}, & \text{if } \rho.c \in \mathcal{C} \\ \displaystyle\sum_{l=2}^{L} \frac{\mathbf{v}_l^\rho \cdot \mathbf{z}_{\text{bg}}^T}{\|\mathbf{v}_l^\rho\| \cdot \|\mathbf{z}_{\text{bg}}\|}, & \text{if } \rho.c \in \mathcal{C}_{\text{bg}} \end{cases} \tag{8}$$

where $\mathbf{z}_c$ and $\mathbf{z}_{\text{bg}}$ represent the foreground and background unified language features, respectively. L denotes the number of features extracted by scale-aware feature alignment. The dot product measures the similarity between the corresponding features, while the normalization ensures scale invariance.

In the training process, we define the ECFD module loss for a single image as follows:

$$\mathcal{L}_{\text{ECFD}} = \frac{1}{N}\sum_{\rho\in P}^{N} l(\mathbf{z}, \mathbf{v}^\rho) \tag{9}$$

where P represents the set of proposals for the image, and N is the total number of proposals. This distillation process effectively prevents degradation of small object features across network depths while leveraging explicit foreground–background disentanglement to suppress background noise interference on small objects.

D. Rotation-Invariant Feature Distillation

Conventional convolutional kernels guided by standard object detection losses are highly susceptible to rotational variations, particularly when processing small-rotated objects. This vulnerability arises due to their inherent limitations

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in capturing rotation-invariant features. While methods such as enhancing rotated RoI extraction or rotated anchors via multiscale feature fusion [4], [5], [16] partially alleviate rotational sensitivity, they merely supplement deep features with shallow ones. However, this approach fundamentally fails to adequately guide backbone features toward class-specific (angle-agnostic) representations. The root cause is that conventional architectures prioritize spatial localization over rotation-invariant feature learning.

In contrast, unified language features generated through contrastive alignment inherently exhibit rotation invariance to angular variations. Furthermore, they are enriched with class-specific semantic knowledge. This is achieved by decoupling geometric transformations from semantic representation learning during the alignment process. Inspired by this mechanism, we propose RIFD to explicitly steer the generation of class-specific representations. This is achieved by distilling rotation-invariant knowledge from VLM into the detector backbone. To mitigate the negative impact of cross-depth distillation on rotation-invariant feature learning, we constrain distillation to a single depth layer. For a given depth d, the distillation bridge features $\mathbf{v}^d$ are defined as

$$\mathbf{v}^d = \left\{\mathbf{v}_1^d, \mathbf{v}_2^d, \ldots, \mathbf{v}_N^d\right\} \tag{10}$$

where each $\mathbf{v}_i^d$ is computed via (4), i corresponds to the ith proposal index, and N denotes the total number of proposals retained after NMS.

Then, the rotation-invariant distillation matrix $\mathcal{M}(\mathbf{v}^d, \mathbf{z})$ is formulated by aligning the distillation bridge features $\mathbf{v}^d$ with the unified language features $\mathbf{z}$ from (6). Specifically, it is defined as

$$\mathcal{M}\left(\mathbf{v}^d, \mathbf{z}\right) = \begin{bmatrix} \mathbf{v}_1^d \cdot \mathbf{z}^T \\ \mathbf{v}_2^d \cdot \mathbf{z}^T \\ \vdots \\ \mathbf{v}_N^d \cdot \mathbf{z}^T \end{bmatrix} \tag{11}$$

$$\mathbf{v}_i^d \cdot \mathbf{z}^T = \underbrace{\left[\frac{\mathbf{v}_i^d \cdot \mathbf{z}_{c1}^T}{\|\mathbf{v}_i^d\| \cdot \|\mathbf{z}_{c1}\|}, \frac{\mathbf{v}_i^d \cdot \mathbf{z}_{c2}^T}{\|\mathbf{v}_i^d\| \cdot \|\mathbf{z}_{c2}\|}, \cdots \frac{\mathbf{v}_i^d \cdot \mathbf{z}_{\text{bg}}^T}{\|\mathbf{v}_i^d\| \cdot \|\mathbf{z}_{\text{bg}}\|}\right]}_{|\mathcal{C} \cup \mathcal{C}_{\text{bg}}|} \tag{12}$$

where $\mathbf{z} \in \mathbb{R}^{|\mathcal{C} \cup \mathcal{C}_{\text{bg}}| \times D}$, $\mathbf{v}_i^d \in \mathbb{R}^{1 \times D}$, and $\mathcal{M}(\mathbf{v}^d, \mathbf{z}) \in \mathbb{R}^{N \times |\mathcal{C} \cup \mathcal{C}_{\text{bg}}|}$ is the distillation matrix, D denotes the embedding dimension of VLM. The RIFD loss for a single image is defined as

$$\mathcal{L}_{\text{RIFD}} = \sum_{d=2}^{L} \mathcal{L}\left(\mathbf{v}^d, \mathbf{z}, \mathbf{y}\right) \tag{13}$$

$$\mathcal{L}\left(\mathbf{v}^d, \mathbf{z}, \mathbf{y}\right) = \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}\left(\mathbf{v}_i^d, \mathbf{z}, \mathbf{y}\right) \tag{14}$$

$$\mathcal{L}\left(\mathbf{v}_i^d, \mathbf{z}, \mathbf{y}\right) = -\sum_{c \in |\mathcal{C} \cup \mathcal{C}_{\text{bg}}|} \mathbf{y}_{i,c} \cdot \log\left(\frac{\exp\left(\mathcal{M}\left[i, c\right]\right)}{\sum_{j \in |\mathcal{C} \cup \mathcal{C}_{\text{bg}}|} \exp\left(\mathcal{M}\left[i, j\right]\right)}\right) \tag{15}$$

where $\mathbf{y}_{i,c} \in \{0, 1\}$ denotes the one-hot label for the ith proposal's class c, $\mathcal{M}[i, c]$ indicates to the element at the ith row and cth column of the rotation-invariant distillation matrix. The proposed RIFD module addresses rotational ambiguity in object detection by distilling rotation-invariant knowledge from VLM into the detector backbone. By constraining distillation to a single depth layer, it suppresses cross-depth interference, enabling robust class-specific feature learning and achieving strong performance under angle variations.

IV. Experiments

A. Datasets

DOTA [47] is a widely adopted large-scale benchmark for oriented object detection. It is extensively used to evaluate detection algorithms in complex real-world scenarios. Comprising 2806 high-resolution images with resolutions ranging from 800 × 800 to 20 000 × 20 000 pixels, DOTA features objects of diverse scales, orientations, and shapes. For rigorous evaluation of small-rotated object detection, our experiments leverage the **DOTA-v1.5** extension, which notably includes annotations for extremely small instances (less than 10 pixels) omitted in the original version. This expanded dataset includes 16 categories and 403 318 instances, enabling a comprehensive assessment of detection robustness specifically for small-rotated objects across varying scales and orientations. The dataset is partitioned into training, validation, and test splits (1/2, 1/6, and 1/3). The training and validation sets are used for model training and hyperparameter tuning, while the test set is reserved for final evaluation. Detection performance is quantitatively assessed via submission of test results to the DOTA benchmark evaluation server, reporting metrics such as mean average precision (mAP).

HRSC2016 [48] is a high-resolution remote sensing benchmark specifically designed for oriented ship detection, offering a specialized evaluation platform for maritime ship recognition in complex environments. It contains 1061 images with resolutions ranging from 300 × 300 to 1500 × 900 pixels, captured in diverse maritime environments. The dataset is partitioned into training (436 images), validation (181 images), and test (444 images) splits, containing a total of 2976 annotated ship instances. We train our model using the combined training and validation splits and reserve the test split for evaluation, employing AP_{50} and AP_{75} as the primary metrics for assessing detection performance.

B. Implementation Details

Our experiments are conducted on a single NVIDIA RTX 4090 GPU with 24-GB VRAM. The experimental results are produced on the OBBDetection [16] platform. Unless otherwise specified, the detector's backbone is the ResNet50 [49] pretrained on ImageNet [50]. The distillation module employs CLIP with a ResNet50 visual encoder as the teacher model, while the unified language features are generated by its text encoder. The prompt templates for distillation are generated by GPT-4o [51] and manually curated for quality. For each image, our distillation module generates 2000 proposals, from which we sample 36 RoIs to serve as distillation carriers. We employ the SGD optimizer with an initial learning rate of 0.01, momentum of 0.9, and weight decay of 0.0001 to update model parameters during training. To stabilize training, we adopt a cosine annealing learning rate schedule with a linear warm-up phase. The warm-up spans 2096 iterations,

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TABLE I
ABLATION STUDY ON CROSS-DEPTH DISTILLATION WITH DIFFERENT DEPTH CONFIGURATIONS

Group	Layer-C_2	Layer-C_3	Layer-C_4	Layer-C_5	mAP (%)
1	-	-	-	-	66.04
2	✓				66.32
3		✓			66.43
4			✓		65.99
5				✓	65.82
6	✓	✓			66.02
7			✓	✓	66.81
8	✓	✓	✓		66.14
9		✓	✓	✓	66.70
10	✓	✓	✓	✓	**67.42**

TABLE II
ABLATION STUDY ON THE TRADEOFF BETWEEN DUAL DISTILLATION LOSSES

α_1	α_2	mAP (%)
-	-	66.04
0	0.1	66.87
0.05	0	66.76
0.05	0.05	66.48
0.1	0.1	66.63
0.1	0.05	66.14
0.05	0.1	**67.42**
0.5	0.5	62.98
1.0	1.0	59.60
0.5	1.0	62.26
1.0	0.5	60.48

during which the learning rate increases linearly from 0.1% of the initial value (i.e., warmup_ratio = 0.001). The minimum learning rate is set to 1e−6 to ensure sustained parameter updates in late training stages.

Different datasets undergo distinct preprocessing pipelines, and the training configurations are adjusted accordingly. For the DOTA dataset, we adopt a single-scale training configuration paired with a preprocessing strategy that crops original images into 1024 × 1024 patches with a stride of 824 pixels, resulting in a 200-pixel overlap between adjacent patches. Training proceeds with a batch size of 4 for 12 epochs, following the learning rate strategy described above. During image patch merging, the ploy NMS threshold is kept consistent with the configuration in OBBDetection. For the HRSC2016 dataset, we follow the default configuration to adjust image dimensions while maintaining the original aspect ratio. Specifically, the shorter sides of images are resized to 800 pixels, and the longer sides are constrained to a maximum of 1333 pixels. To accommodate dataset-specific characteristics, we adapt the distillation module's weightings during training and modify the learning rate schedule to stabilize training dynamics. Specifically, we employ a step-based learning rate decay with linear warm-up: the learning rate starts at 0.1% of the initial value and increases linearly over 500 iterations during the warm-up phase. After warm-up, the learning rate is reduced by a factor of 0.1 at the 27th and 34th epochs following predefined step milestones.

C. Ablation Analysis

In this section, we conduct a comprehensive ablation study on the DOTA-v1.5 test set to evaluate the effectiveness of our proposed method. Unless specified otherwise, the original Oriented R-CNN serves as the baseline model.

1) Cross-Depth Distillation Configuration: In scenarios where both the detector backbone and the visual encoder in the distillation module use ResNet50, we verify the effectiveness of cross-depth distillation. Moreover, we determine the optimal configuration for the distillation depth. As shown in Table I, we evaluate the effectiveness of cross-depth distillation across four configurations: single-depth (groups 2–5), dual-depth (groups 6 and 7), triple-depth (groups 8 and 9), and quadruple-depth (group 10). Here, Layer-C_i indicates the distillation bridge features derived from the C_ith layer of the detector's backbone. Each group represents a distinct configuration of these depths, with ✓ indicating active feature usage in cross-depth distillation.

Analysis of single-depth configurations reveals a striking performance dichotomy depending on the distilled layer depth. When distilling lower level features (i.e., C_1–C_2), the detector achieves mAP improvements of 0.42–0.59% over the baseline, demonstrating enhanced capability in capturing small-rotated objects. In contrast, distilling higher level features (i.e., C_3–C_4) results in performance degradation, falling below the baseline. This dichotomy suggests that the distillation module effectively reinforces spatial details and angle-agnostic representations encoded in lower layers, which are critical for localizing small objects with rotational variations. However, higher level features suffer from semantic abstraction, resulting in the loss of fine-grained geometric precision. Distilling only high-level features fails to guide lower layers due to this loss of detail. In addition, the distributional mismatch between distilled high-level and undistilled low-level features exacerbates optimization challenges, further limiting detection performance for small-rotated objects. Due to FPN's reliance on multiscale features from all backbone layers for downstream tasks, configurations omitting any layer inevitably suffer from distributional mismatches. Experimental validation in Table I confirms this: the quadruple-depth distillation (group 10), which leverages all four backbone layers, achieves the highest mAP of 67.42%, outperforming configurations with fewer layers.

2) Dual Loss Tradeoff in Distillation: As a plug-and-play module, our framework implements the dual loss tradeoff mechanism in distillation by formulating the detector's end-to-end optimization objective as

$$\mathcal{L}_{\text{overall}} = \mathcal{L}_{\text{det}} + \alpha_1 \mathcal{L}_{\text{ECFD}} + \alpha_2 \mathcal{L}_{\text{RIFD}} \tag{16}$$

where α_1 and α_2 govern the tradeoff between the base detection loss $\mathcal{L}_{\text{det}}$ and the distinct distillation components. To further analyze the dual loss tradeoff, we conduct ablation studies on the hyperparameters α_1 and α_2 under the optimal quadruple-depth configuration, thereby assessing their impact on detection performance. We systematically explore various combinations of $\{\alpha_1, \alpha_2\}$ in the range of 0 to 1.0 to determine the optimal parameter settings. Table II shows that the baseline (no distillation) achieves 66.04% mAP. Enabling either loss

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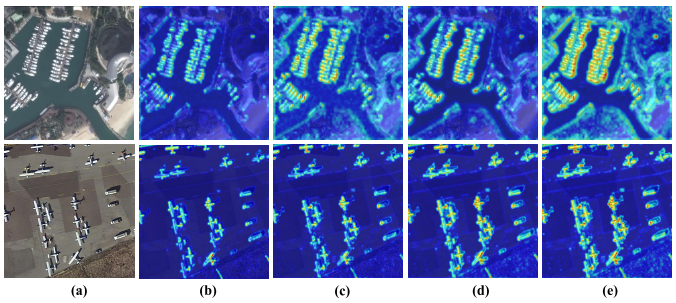


Fig. 4. Qualitative results on the contributions of ECFD and RIFD. We select two representative scenes: harbor ships, which suffer from feature dilution, and airport airplanes, which exhibit orientation-dependent feature quality. These cases allow qualitative validation of each module's effectiveness on its targeted problem and their complementary benefits when combined. Feature maps are extracted from different depths of the backbone: the third block for the harbor scene, as feature dilution becomes more pronounced at deeper layers due to background noise, and the first block for the airport scene, as insufficient rotation invariance already emerges in high-resolution shallow features. (a) Source. (b) Baseline. (c) +ECFD. (d) +RIFD. (e) CRKD.

alone yields marginal gains: $\{0.05, 0\}$ improves mAP by + 1.09%, while $\{0, 0.1\}$ achieves a + 1.26% boost. The optimal performance (67.42% mAP) is achieved with $\alpha_1 = 0.05$ and $\alpha_2 = 0.1$, outperforming all other configurations by + 0.82%–1.94%. This result underscores the synergistic effect of combining both losses, where ECFD mitigates feature dilution for small objects and RIFD enhances rotation invariance.

Qualitative results in Fig. 4 further highlight the independent contributions and complementarity of the two losses. In the harbor scene, baseline outputs suffer severe dilution: background noise suppresses most left-side ships, while central ships retain only head responses with missing body structures. ECFD effectively recovers the complete structural features of central ships and restores the left-side ships, demonstrating its ability to mitigate depth-induced feature dilution. In contrast, RIFD emphasizes class-dominant, rotation-invariant semantics at a single depth. It enhances the left-side ships and maintains consistency with the central ships, but cannot recover diluted body responses—reinforcing ship heads while leaving missing structures absent. In the airport scene, baseline features show inconsistent quality across airplanes with different orientations, reflecting insufficient rotation invariance. ECFD introduces little improvement, as dilution is not the dominant issue. Conversely, RIFD significantly strengthens low-quality airplane features and unifies their representations across orientations, thereby improving rotation robustness. When combined, ECFD and RIFD (**CRKD**) complement each other by jointly addressing feature dilution and insufficient rotation invariance. As shown in Fig. 4, this synergy enables small-rotated objects to preserve complete structures while maintaining rotation-invariant semantics, producing more discriminative and angle-agnostic representations.

Moreover, RIFD produces more consistent and complete airplane features across different orientations, illustrating that language supervision functions as a rotation-invariant semantic reference in 2-D remote sensing detection. This evidence supports our earlier claim that language embeddings from VLMs inherently encode orientation-independent semantics.

TABLE III
ABLATION STUDY ON THE LOSS CONFIGURATIONS FOR DUAL DISTILLATION

$\mathcal{L}_{\text{ECFD}}$	$\mathcal{L}_{\text{RIFD}}$	mAP (%)
Smooth L1 Loss	Cross-Entropy Loss	66.55
Smooth L1 Loss	KL Divergence	66.29
MSE	Cross-Entropy Loss	66.62
MSE	KL Divergence	66.27
Cosine Similarity	Cross-Entropy Loss	**67.42**
Cosine Similarity	KL Divergence	66.49

Since textual descriptions in VLM training lack explicit directional cues, the model learns category-level semantics that remain stable under rotation. Such angle-agnostic textual embeddings provide consistent guidance for visual feature alignment, enabling RIFD to restore discriminative cues lost due to orientation variations.

3) Loss Configurations for Dual Distillation: To further explore the impact of loss function selection on our dual distillation framework, we evaluate different combinations of $\mathcal{L}_{\text{ECFD}}$ and $\mathcal{L}_{\text{RIFD}}$ to optimize feature consistency and rotation invariance. The results in Table III reveal that the choice of loss functions significantly influences the tradeoff between preserving spatial details and resolving rotational ambiguity. The optimal configuration is achieved when $\mathcal{L}_{\text{ECFD}}$ employs cosine similarity loss and $\mathcal{L}_{\text{RIFD}}$ uses cross-entropy loss. This combination aligns with the design principles of ECFD and RIFD: cosine similarity enforces consistency across multidepth detector features, preserving fine-grained spatial structures critical for small objects. In contrast, cross-entropy loss for RIFD provides a robust optimization landscape for classifying rotation-invariant features derived from the distillation matrix $\mathcal{M}(\mathbf{v}^d, \mathbf{z})$ [see (11)], ensuring explicit semantic disambiguation between categories.

The effectiveness of cosine similarity for ECFD stems from its ability to align geometrically consistent features across network depths while suppressing background noise. As shown in (8), this loss measures angular similarity between detector features and unified language features, directly addressing the foreground–background disentanglement objective. In contrast, losses like smooth L1 or MSE prioritize magnitude matching over structural preservation, leading to suboptimal performance (66.27%–66.62% mAP) as they fail to capture the spatial relationships critical for cross-depth alignment. For RIFD, cross-entropy loss outperforms KL divergence due to its stability in handling class-specific rotation-invariant learning. The RIFD formulation [see (13)] requires explicit classification of proposals into categories (including background), where the log-likelihood formulation of the cross-entropy provides a clear optimization signal. KL divergence, while effective for distribution matching, introduces instability in the rotation-invariant context due to its sensitivity to outlier samples—a limitation exacerbated by the angular variability in remote sensing scenes.

The synergy between cosine similarity and cross-entropy reflects their complementary roles: the former ensures geometric consistency across detector depths to counteract feature dilution, while the latter enforces discriminative rotation

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TABLE IV
ABLATION STUDY ON VLM-AS-BACKBONE AND VLM-DISTILLATION

Model	mAP (%)	Params (M)
VLM-as-Backbone		
CLIP-R50 [12]	54.09	41.17
RemoteCLIP-R50 [52]	56.76	41.17
CLIP-R101 [12]	57.35	60.16
VLM-Distillation		
RemoteCLIP-R50 → R50	66.84	41.14
CLIP-R101 → R50	66.93	41.14
CLIP-R50 → R50 (CRKD)	**67.42**	**41.14**

invariance by leveraging the structured semantics of unified language features. This configuration aligns with our design choices for ECFD and RIFD, which prioritize spatial fidelity and class-specific rotation robustness, respectively. These results underscore the necessity of characteristic-aware loss selection in rotation detection tasks. By tailoring $\mathcal{L}_{\text{ECFD}}$ to preserve spatial details through cross-depth alignment and $\mathcal{L}_{\text{RIFD}}$ to resolve rotational ambiguity via robust classification, our framework maximizes the synergy between KD and detection performance, forming the final optimized configuration for our method.

4) VLM-as-Backbone Versus VLM-Distillation: To validate the necessity of our distillation approach, we compare VLM-as-backbone and VLM-distillation strategies. As shown in Table IV, directly using VLMs (CLIP-R50, RemoteCLIP-R50 [52], and CLIP-R101) as backbones for detection yields significantly inferior performance (54.09%–57.35% mAP) compared to our distillation method (67.42% mAP), despite similar model complexity (41.14 versus 41.17 M params). This performance gap stems from a fundamental mismatch between VLM pretraining objectives and detection task requirements. VLMs are designed to prioritize text-prompted global semantic alignment, which does not align with standard object detection pipelines that lack prompt inputs. Consequently, their direct adoption as backbones fails to support precise spatial localization of instances without prompt guidance, which is essential for small-rotated object detection. In remote sensing, detecting small objects depends on extracting semantics from extremely local regions, which conflicts with the global semantic paradigm underlying VLMs. We therefore exploit the cross-modal alignment capacity of VLMs to enhance instance-level representations, explicitly enforcing semantic consistency and rotation invariance crucial for small-rotated object detection.

Quantitatively, CRKD (CLIP-R50 →R50) achieves the best performance (67.42% mAP), outperforming all VLM-as-backbone approaches by 17.56%–24.64%. The flexibility of our framework is further demonstrated by two key experiments: CLIP-R101 →R50 (66.93% mAP) demonstrates effective knowledge transfer from larger VLMs to smaller backbones without computational overhead. RemoteCLIP-R50 →R50 (66.84% mAP) showcases the capability to customize distillation teachers for domain-specific requirements, both while maintaining inference efficiency. Our success lies in utilizing VLMs for instance-level alignment rather than global image processing. By focusing on instance features instead of whole-image features, CRKD successfully preserves spatial details while enforcing rotation-invariant representations, addressing the core challenges of small-rotated object detection.

5) Distillation Across Detection Frameworks: To validate the cross-framework efficacy of our proposed distillation approach, we evaluate its performance across diverse oriented object detection frameworks. Table V demonstrates consistent performance improvements across six architectures, including single-stage (Rotated RetinaNet [53], R^3Det, and S^2ANet) and two-stage (Rotated Faster R-CNN [54], Gliding Vertex [55], and Oriented R-CNN) detectors. These results validate CRKD's adaptability to heterogeneous architectures while addressing critical challenges in detecting small-rotated objects, as evidenced by significant gains in categories like small vehicle (SV) and helicopter (HC).

Quantitatively, CRKD achieves notable improvements in both detection accuracy and category-specific robustness. For single-stage methods, Rotated RetinaNet benefits most, with mAP increasing from 59.16% to 61.61% (+4.14%). This improvement is driven by enhanced performance in challenging categories: SV rises from 44.53% to 50.10%, reflecting ECFD's ability to preserve discriminative features for small objects. PL improves from 71.43% to 78.22%, demonstrating RIFD's role in reducing class confusion for rotation-sensitive objects. SH, a cluttered small-scale category, achieves a 6.93% boost (from 73.31% to 78.39%), highlighting the effectiveness of ECFD in preserving spatial coherence across network depths. In two-stage frameworks, Oriented R-CNN shows the largest overall improvement, with mAP rising from 66.04% to 67.42% (+ 2.09%). This aligns with our ablation studies and underscores CRKD's effectiveness for complex architectures: CC, a category with high rotational ambiguity, improves from 5.03% to 18.71%, illustrating RIFD's ability to disentangle class semantics from orientation-dependent features. HC gains + 13.26% (from 53.01% to 60.04%), while SH improves from 80.85% to 86.50%, demonstrating synergistic benefits for both small-scale and rotated objects.

The cross-framework effectiveness of CRKD is further substantiated by analyzing the predistillation and postdistillation scenarios in Fig. 5. In the parking lot scene, the baseline exhibits a pronounced missed detection problem for SVs in clustered regions. This failure stems from the progressive feature dilution caused by deep networks, where high-frequency texture details—critical for distinguishing compact objects—are suppressed during feature extraction. The resulting dominance of global context over local patterns leads to false negatives, as evidenced by the absence of bounding boxes in shadowed or clustered areas. By enforcing cross-depth consistency, ECFD preserves fine-grained texture information across pyramid levels, enabling robust detection even in low-texture or clustered regions. This is reflected in the postdistillation results, where clustered SVs are reliably localized without introducing false positives, demonstrating the efficacy of the module in counteracting feature dilution. In the airport runway scenario, the baseline suffers from

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TABLE V

PERFORMANCE COMPARISON OF **CRKD** ACROSS SINGLE-STAGE AND TWO-STAGE DETECTION FRAMEWORKS. THE TABLE REPORTS MAP (%) AND CATEGORY-WISE METRICS FOR BASELINE MODELS AND THEIR VARIANTS WITH CRKD

Method	Config	mAP	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	CC
Single-stage methods																		
Rotated RetinaNet [53]	baseline	59.16	71.43	77.64	42.12	64.65	44.53	56.79	73.31	90.84	76.02	59.96	46.95	69.24	59.65	64.52	48.06	0.83
	+CRKD	**61.61**	78.22	79.67	40.34	64.93	50.10	64.03	78.39	90.82	76.95	71.10	52.72	70.81	56.48	63.95	47.17	0.00
R^3Det [5]	baseline	61.24	78.37	76.94	45.73	66.80	48.90	70.14	79.71	90.86	74.08	69.53	49.85	70.14	61.71	63.61	33.55	0.00
	+CRKD	**62.61**	78.68	75.59	46.14	69.04	49.08	71.75	79.87	90.88	76.31	68.59	53.73	70.49	63.76	63.54	44.32	0.00
S^2ANet [4]	baseline	64.29	79.24	77.77	50.03	69.59	56.51	73.80	80.63	90.88	76.98	75.59	50.01	74.58	64.00	63.36	45.60	0.00
	+CRKD	**65.55**	78.75	76.44	50.55	69.80	60.80	76.28	86.88	90.86	77.11	76.06	53.89	72.37	66.31	66.78	45.89	0.00
Two-stage methods																		
Rotated Faster R-CNN [54]	baseline	62.00	71.89	74.47	44.45	59.87	51.28	68.98	79.37	90.78	77.38	67.50	47.75	69.72	61.22	65.28	60.47	1.54
	+CRKD	**64.66**	79.92	79.87	46.45	65.54	51.76	68.47	79.28	90.86	78.37	68.25	60.13	73.42	57.14	65.63	63.67	5.74
Gliding vertex [55]	baseline	63.83	72.25	75.74	49.14	65.67	51.83	73.64	80.58	90.82	78.67	67.94	53.65	72.33	65.34	65.57	52.16	5.95
	+CRKD	**65.22**	78.03	79.58	51.15	66.18	51.88	73.58	80.53	90.80	78.36	68.73	60.90	72.22	64.56	64.69	61.31	1.02
Oriented R-CNN [16]	baseline	66.04	79.20	79.75	53.36	71.20	52.05	76.07	80.85	90.85	81.88	68.12	60.75	71.81	67.30	65.36	53.01	5.03
	+CRKD	**67.42**	79.99	78.93	53.12	68.27	52.18	76.02	86.50	90.84	77.83	68.47	60.18	72.48	70.93	64.17	60.04	18.71

TABLE VI

COMPARISONS WITH THE STATE-OF-THE-ARTS ON THE DOTA-V1.5 DATASET. THE SUPERSCRIPT * DENOTES METHODS THAT ADOPT THE OVD PIPELINE

Method	mAP	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	CC
Rotated RetinaNet [53]	59.16	71.43	77.64	42.12	64.65	44.53	56.79	73.31	90.84	76.02	59.96	46.95	69.24	59.65	64.52	48.06	0.83
Rotated Faster R-CNN [54]	62.00	71.89	74.47	44.45	59.87	51.28	68.98	79.37	90.78	77.38	67.50	47.75	69.72	61.22	65.28	60.47	1.54
R^3Det [5]	61.24	78.37	76.94	45.73	66.80	48.90	70.14	79.71	90.86	74.08	69.53	49.85	70.14	61.71	63.61	33.55	0.00
S^2ANet [4]	64.29	79.24	77.77	50.03	69.59	56.51	73.80	80.63	**90.88**	76.98	**75.59**	50.01	**74.58**	64.00	63.36	45.60	0.00
Gliding vertex [55]	63.83	72.25	75.74	49.14	65.67	51.83	73.64	80.58	90.82	78.67	67.94	53.65	72.33	65.34	65.57	52.16	5.95
Oriented Reppoints [56]	61.80	75.15	72.04	49.40	67.86	56.40	71.79	80.42	90.88	75.84	74.39	42.19	69.51	65.15	63.26	34.46	0.10
Mask R-CNN-O [57]	62.67	76.84	73.51	49.90	57.80	51.31	71.34	79.75	90.46	74.21	66.07	46.21	70.61	63.07	64.46	57.81	9.42
HTC-O [58]	63.40	77.80	73.67	51.40	63.99	51.54	73.31	80.31	90.48	75.12	67.34	48.51	70.63	64.84	64.48	55.87	5.15
RoI Transformer [15]	66.60	72.30	**81.83**	52.40	71.09	52.44	75.54	80.94	90.88	**82.47**	68.91	**62.84**	73.67	68.41	66.58	59.66	5.68
SASM [17]	58.98	71.49	66.93	46.94	64.67	52.09	69.66	76.08	90.82	71.46	72.92	40.08	68.37	62.05	60.55	29.62	0.01
Oriented R-CNN [16]	66.04	79.20	79.75	**53.36**	**71.20**	52.05	**76.07**	80.85	90.85	81.88	68.12	60.75	71.81	67.30	65.36	53.01	5.03
DFDet [59]	66.46	77.62	80.56	48.14	69.42	**62.48**	75.27	86.14	90.84	77.99	75.40	50.48	70.04	64.48	**72.46**	51.61	10.43
CastDet* [60]	57.67	77.15	66.92	42.18	54.64	50.65	70.91	79.81	89.47	74.70	68.49	41.33	62.47	55.47	65.25	28.16	1.05
Oriented ViLD* [61]	59.21	71.91	70.86	43.31	58.70	50.83	72.01	79.01	90.74	73.06	67.49	37.55	65.83	48.85	63.37	53.79	0.10
CRKD	**67.42**	**79.99**	78.93	53.12	68.27	52.18	76.02	**86.50**	90.84	77.83	68.47	60.18	72.48	**70.93**	64.17	**60.04**	**18.71**

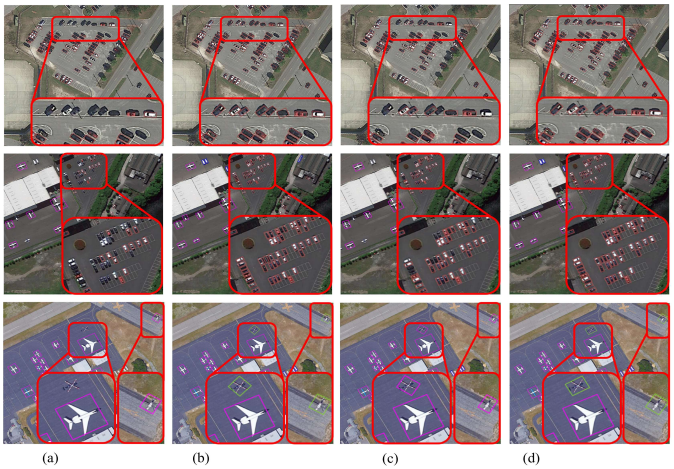


Fig. 5. Qualitative comparisons between baseline methods with our proposed CRKD distilled methods on the DOTA-v1.5 dataset. (a) R^3Det. (b) CRKD + R^3Det. (c) O-RCNN. (d) CRKD + O-RCNN.

misclassification, particularly for HCs with extreme orientations. These errors stem from the rotation sensitivity of convolutional kernels, which can map different classes to similar embeddings when their orientations align. In particular, because geometric transformations remain entangled with class-specific features, HCs whose headings approach those of airplanes are occasionally labeled as planes. RIFD addresses this by leveraging the inherent invariance of unified language features to angular variations. By constraining distillation to a single depth layer, RIFD avoids cross-depth interference. The resulting rotation-invariant distillation matrix $\mathcal{M}(\mathbf{v}^d, \mathbf{z})$ decouples geometric transformations from class semantics, ensuring robust classification under angular variations. This is evident in the postdistillation results, where misaligned bounding boxes are corrected, and rotation-sensitive HCs are accurately classified, highlighting the ability of RIFD to resolve rotation-dependent feature conflicts. These findings confirm that CRKD systematically addresses feature dilution and rotation dependency—through explicit language-guided alignment and rotation-invariant knowledge transfer, achieving cross-framework generalization without architectural modifications.

D. Comparison With the State-of-the-Art

We evaluate CRKD against state-of-the-art oriented object detection methods on two challenging benchmarks: the DOTA-v1.5 dataset (multicategory rotated objects) and the HRSC2016 dataset (small maritime objects). Tables VI and VII demonstrate the superiority of CRKD in addressing critical challenges of small-scale and rotated object detection through its dual-distillation framework.

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TABLE VII
COMPARISONS WITH THE STATE-OF-THE-ART METHODS ON THE HRSC2016 DATASET

Method	AP_{50} (%)	AP_{75} (%)
RoI Transformer [15]	90.05	54.43
Oriented R-CNN [16]	93.46	81.41
Gliding Vertex [55]	83.26	31.88
S^2ANet [4]	93.03	54.85
Oriented RepPoints [56]	91.92	66.04
SASM [17]	92.44	72.89
CastDet* [60]	92.60	77.60
Oriented ViLD* [61]	80.40	52.60
CRKD	**94.93**	**81.93**

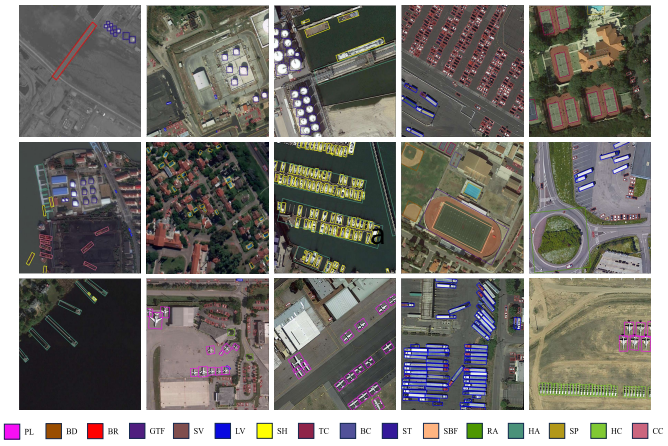


Fig. 6. Examples of detection results on the DOTA-v1.5 dataset using our CRKD with ResNet50 backbone, where the confidence threshold is set to 0.3 when visualizing these results.

1) Results on DOTA-v1.5 Dataset: CRKD achieves an mAP of 67.42%, outperforming advanced methods such as DFDet (66.46%), RoI Transformer (66.60%), and SASM (58.98%). Its performance improvements are most pronounced in categories requiring robustness to rotation and scale variations. For CC, a category with extreme rotational ambiguity, CRKD achieves 18.71% mAP, far surpassing Mask R-CNN-O (9.42%), DFDet (10.43%), and Gliding Vertex (5.95%). In SV, CRKD reaches 52.18% mAP, outperforming most existing methods, demonstrating its ability to preserve discriminative features for small objects. For HC, CRKD's 60.04% mAP exceeds RoI Transformer (59.66%) and S^2Net (45.60%), highlighting reduced class confusion in rotation-sensitive categories. For medium-to-large-scale categories like SBF, CRKD almost surpasses all methods, outperforming the second-best method (Gliding Vertex at 53.65%) by + 12.17% with 60.18% mAP. In HA, CRKD achieves 70.93% mAP, exceeding all other methods. These results underscore an additional advantage of CRKD: its cross-depth feature enhancement not only preserves discriminative cues for small objects but also improves spatial coherence for medium-to-large-scale objects. Meanwhile, the rotation-invariant learning of RIFD further reduces ambiguities in complex, cluttered scenes across all object sizes. We also compare CRKD with two representative VLM-based methods, CastDet [60] and Oriented ViLD [61], which follow the open-vocabulary detection

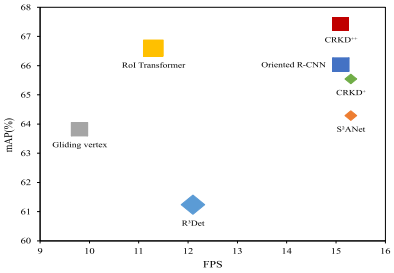


Fig. 7. Efficiency comparison of state-of-the-art methods on the DOTA-v1.5 dataset in terms of accuracy (mAP), floating-point operations (FLOPs), and inference speed (FPS). Each point is categorized as a single-stage (diamonds) or two-stage (squares) detectors through distinct geometric shapes, while color encodes individual methods (e.g., orange for S^2ANet). The area of each marker correlates with their computational cost (FLOPs), where larger markers indicate higher complexity. Notably, CRKD$^+$ denotes the distillation variant built upon the single-stage baseline S^2ANet, whereas CRKD^{++} represents the distillation version based on the two-stage Oriented R-CNN framework.

(OVD) pipeline. To ensure a fair comparison, we adopt the same dataset preprocessing as MMRotate [62] and follow the experimental settings of CastDet [60]. As shown in Table VI, CastDet and Oriented ViLD achieve 57.67% and 59.21% mAP, both notably lower than CRKD's 67.42%. Their performance drops significantly in rotation-sensitive categories (e.g., HC: 28.16% and 53.79%; CC: 1.05% and 0.10%), revealing limited robustness to geometric variations. Although textual supervision in VLMs is inherently rotation-invariant, these OVD pipelines lack effective mechanisms to guide such invariance into instance-level visual representations. In contrast, CRKD explicitly distills rotation-invariant and scale-consistent knowledge, achieving stronger discriminability for both small and rotated objects. In Fig. 6, we give some detection examples on the DOTA-v1.5 dataset, where our method can detect small-rotated objects even under challenging conditions.

2) Results on HRSC2016 Dataset: On this ship detection dataset, CRKD achieves 94.93% AP_{50} and 81.93% AP_{75}, surpassing all methods. AP_{50} prioritizes object presence over bounding box precision, evaluating the detector's ability to detect small, easily missed objects. CRKD surpasses S^2ANet by + 2.04% and Oriented R-CNN by + 1.57% in AP_{50}, demonstrating superior robustness in localizing tiny ships amid severe scale variation. This gain arises from our ECFD module, which preserves discriminative features across depths to counteract small object degradation. AP_{75} demands precise orientation and spatial alignment, where minor angular deviations can drastically reduce IoU. CRKD achieves 81.93% AP_{75}, outperforming SASM by + 12.40% and Oriented R-CNN by + 0.64%, demonstrating resistance to diverse ship orientations. This improvement owes to the RIFD module, which decouples rotation-invariant features and ensures accurate localization under strict IoU constraints. A unified architecture ensures consistent performance across scales and orientations. By integrating the small object consistency of the ECFD module with the rotation-invariant learning of the RIFD module, CRKD simultaneously addresses the dual challenges of detecting small-rotated ships in cluttered scenes.

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3) Computational Efficiency Analysis: We demonstrate that CRKD achieves state-of-the-art performance on the DOTA-v1.5 and HRSC2016 benchmarks while maintaining computational efficiency (see Fig. 7). Specifically, CRKD matches the baseline in FLOPs and inference speed, introducing zero additional computational overhead during inference. This is a critical advantage over competing methods such as RoI Transformer and Gliding Vertex, which employ complex modules like transformer-based attention or iterative vertex refinement. These design choices increase computational load without proportional accuracy gains. The efficiency of CRKD is rooted in its architectural design philosophy: the dual distillation framework integrates seamlessly into the backbone architecture without introducing new learnable parameters or forward-pass operations during inference. From a numerical perspective, the FLOPs of CRKD align with those of the baseline, reflecting their shared detector structure. However, CRKD outperforms the baseline in mAP by leveraging KD during training without altering the inference pipeline. This design mirrors principles from production efficiency analysis, where resource allocation is optimized to maximize output without increasing input costs.

V. Conclusion

In this article, we present a novel framework, CRKD, which leverages VLMs to address critical challenges in RSOD, particularly focusing on small-rotated objects. By introducing ECFD and RIFD, we effectively mitigate feature dilution and enhance rotation invariance, respectively. Our methods ensure robust feature representation across multiple network depths and orientations, significantly boosting detection accuracy. The proposed framework not only surpasses state-of-the-art methods on challenging RSOD benchmarks but also demonstrates broad applicability across different detector architectures. These findings underscore the potential of integrating multimodal representation learning to tackle complex detection scenarios, paving the way for future advancements in visual perception technologies.

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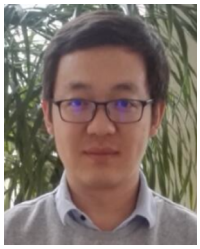
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